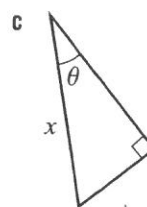
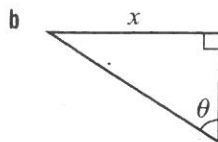
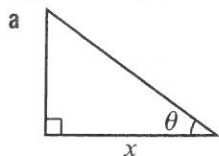


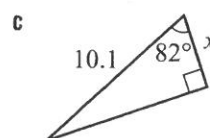
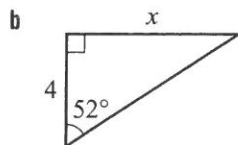
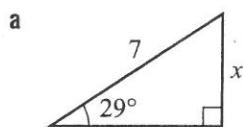
und as

- 1 For the marked angle θ , decide if x represents the length of the opposite (O), adjacent (A) or hypotenuse (H) side.



- 2 Decide if you would use $\sin \theta = \frac{O}{H}$, $\cos \theta = \frac{A}{H}$ or $\tan \theta = \frac{O}{A}$ to help find the value of x in these triangles.

Do not find the value of x , just state which ratio would be used.



(H) are

7(H)



- Example 8 3 Use a calculator to evaluate the following correct to two decimal places.



- a $\sin 20^\circ$ b $\cos 37^\circ$ c $\tan 64^\circ$ d $\sin 47^\circ$
e $\cos 84^\circ$ f $\tan 14.1^\circ$ g $\sin 27.4^\circ$ h $\cos 76.2^\circ$

- Example 9 4 In each of the following, find the value of x correct to two decimal places.



- a $\sin 50^\circ = \frac{x}{4}$ b $\tan 81^\circ = \frac{x}{3}$ c $\cos 33^\circ = \frac{x}{6}$
d $\cos 75^\circ = \frac{x}{3.5}$ e $\sin 24^\circ = \frac{x}{4.2}$ f $\tan 42^\circ = \frac{x}{10}$
g $\frac{x}{7.1} = \tan 18.4^\circ$ h $\frac{x}{5.3} = \sin 64.7^\circ$ i $\frac{x}{12.6} = \cos 52.9^\circ$

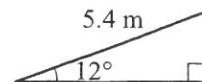
- Example 10 5 For the triangles given below, find the value of x correct to two decimal places.



- a
- b
- c
- d
- e
- f
- g
- h
- i
- j
- k
- l
- m
- n
- o
- p



- 6 Amy walks 5.4 m up a ramp which is inclined at 12° to the horizontal. How high (correct to two decimal places) is she above her starting point?



c no, adjacent side can't be zero

d no, as the numerator < denominator for sin and cos as the hypotenuse is the longest side

15 a $\sin \theta = \frac{3}{5}$ $\tan \theta = \frac{3}{4}$

b i $\cos \theta = \frac{\sqrt{3}}{2}$ $\tan \theta = \frac{1}{\sqrt{3}}$

ii $\tan \theta = \sqrt{3}$ $\sin \theta = \frac{\sqrt{3}}{2}$

iii $\sin \theta = \frac{1}{\sqrt{2}}$ $\cos \theta = \frac{1}{\sqrt{2}}$

c equals one

d $(\sin \theta)^2 + (\cos \theta)^2 = 1$ (the Pythagorean identity)

Exercise 3F

- 1 a A b 0 c H
- 2 a sin b tan c cos
- 3 a 0.34 b 0.80 c 2.05 d 0.73
e 0.10 f 0.25 g 0.46 h 0.24
- 4 a 3.06 b 18.94 c 5.03
d 0.91 e 1.71 f 9.00
g 2.36 h 4.79 i 7.60
- 5 a 5.95 b 0.39 c 13.38 d 3.83
e 8.40 f 1.36 g 29.00 h 1.62
i 40.10 j 4.23 k 14.72 l 13.42
m 17.62 n 5.48 o 9.75 p 1.01
- 6 1.12 m 7 44.99 m 8 10.11 m
- 9 a 20.95 m b 10 cm
- 10 a 65° b 1.69 c 1.69
- 11 a i 80° ii 62° iii 36° iv 9°
b i both 0.173... ii both 0.469...
iii both 0.587... iv both 0.156...
c $\sin \theta^\circ = \cos (90^\circ - \theta^\circ)$
d i 70° ii 31° iii 54° iv 17°
- 12 a $\sqrt{2}$
b i $\frac{1}{\sqrt{2}}$ ii $\frac{1}{\sqrt{2}}$ iii 1 c $\sqrt{3}$
d i $\frac{1}{2}$ ii $\frac{\sqrt{3}}{2}$ iii $\frac{1}{\sqrt{3}}$ iv $\frac{\sqrt{3}}{2}$ v $\frac{1}{2}$ vi $\sqrt{3}$

Exercise 3G

- 1 a $x=2$ b $x=5$ c $x=3$ d $x=4$
e $x=7$ f $x=4$ g $x=0.5$ h $x=0.2$
- 2 a 4.10 b 6.81 c 37.88 d 0.98
e 12.80 f 14.43 g 9.52 h 114.83
i 22.05
- 3 a 13.45 b 16.50 c 57.90 d 26.33
e 15.53 f 38.12 g 9.15 h 32.56
i 21.75 j 49.81 k 47.02 l 28.70